Dear Members of the Admissions Committee,

I am pleased to recommend Hon Li for admission to the Online Master of Science in Analytics program at the Georgia Institute of Technology, with a concentration in Computational Data Analytics. I supervised Hon's work throughout the design, implementation, experimentation, and evaluation of a reinforcement learning project. This experience gave me the opportunity to observe his technical ability, analytical judgment, and potential for advanced study.

The project focused on training an intelligent agent to play Atari Breakout using deep reinforcement learning. Within the team, Hon took primary responsibility for implementing the convolutional neural network, the Deep Q-Network, and the DQN agent. These components formed the technical core of the system. He helped convert raw game observations into a representation suitable for learning by processing four consecutive grayscale frames, allowing the model to infer not only the positions of the ball and paddle but also their movement over time. He then implemented a three-layer convolutional architecture that transformed the image input into a compact feature vector for Q-value estimation.

Hon also demonstrated a sound understanding of the principles behind reinforcement learning rather than treating the model as a black box. His implementation incorporated an experience replay buffer, epsilon-greedy exploration, separate policy and target networks, and soft target-network updates. Integrating these elements into a functioning training pipeline required careful attention to tensor dimensions, batch processing, action selection, and the Bellman update. His ability to connect the mathematical concepts of DQN with a working PyTorch implementation was one of the strongest aspects of his contribution.

Equally important was Hon's approach to experimental analysis. The team trained and evaluated agents under both fixed and randomized environment seeds and compared standard DQN with Double DQN. Hon recognized that the strong performance observed under a fixed seed did not necessarily imply good generalization. He understood that an agent trained repeatedly on one environment configuration could become specialized to that configuration and perform less reliably on unfamiliar trajectories. In examining the DQN and Double DQN results, he also considered stability, Q-value overestimation, and the tradeoff between peak performance and consistency. The team further identified prioritized experience replay as a promising direction for improving learning from rare but consequential transitions. This willingness to question results and identify methodological limitations reflects the kind of analytical maturity required for graduate study.

Hon worked effectively within a three-person team while maintaining clear ownership of substantial technical components. He was able to explain the architecture and experimental results in the team's final report and presentation, helping connect implementation details to the broader questions of model performance and generalization. His work showed persistence in dealing with a computationally demanding problem and intellectual curiosity beyond merely obtaining a high game score.

I believe Hon is well suited to Georgia Tech's Computational Data Analytics track. His experience with machine learning, high-dimensional image data, computational modeling, model evaluation, and experimental comparison provides a strong foundation for its rigorous curriculum. More importantly, he has demonstrated the ability to implement complex methods, interpret their results critically, and learn from their limitations.

I recommend Hon Li with confidence for admission to the Online Master of Science in Analytics program. I believe he will be a capable and thoughtful contributor to the Georgia Tech academic community.

Sincerely,

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